

Cognate redocking as a quality-control step for cryo-EM kinase receptor structures: a study of ATR/CHK1 and CDK-activating kinase

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Abstract Reproducing a receptor's own deposited experimental ligand model (cognate redocking) is a quality check commonly applied in structure-based virtual screening. The cryo-EM ATR structures bound to berzosertib (PDB 9L40, 2.87 Å) and camonsertib (PDB 9L4B, 3.20 Å) were redocked with AutoDock Vina (2.0 Å symmetry-corrected RMSD pass threshold): 9L4B passed (0.61 ± 0.02 Å) while the higher-resolution 9L40 failed (2.53 ± 0.01 Å), not explained by the nearby allosteric pocket (60.65 Å away). The procedure was repeated on 14 cryo-EM structures of the CDK7/cyclin H/MAT1 CAK complex (a single-target series): again, 13/14 structures did not pass. This is not a comprehensive benchmark, but it does serve as a pipeline sanity check: the same pipeline was applied to 20 objectively chosen CASF-2016 complexes (8/20 passing, 40%; exact 95% CI 19–64%), arguing against a badly broken pipeline. Neither nominal resolution nor the wwPDB map/model metrics (Q-score, residue inclusion) correlated with redocking performance; given only two Q-scored passing structures, this is a descriptive finding, not a conclusive statistical correlation. Against a matched rigid control (4.50 ± 0.05 Å), allowing five active-site residues within 4.5 Å of native berzosertib to move improved the pose of 9L40 (3.00 ± 0.92 Å) but did not let it pass. Docking a 40-compound pilot library against excluded 9L40 instead of validated 9L4B left the top five compounds unchanged but reshuffled the screen's mid-ranks (Spearman $\rho = 0.21$; only 5/10 shared top-10 compounds). Cognate redocking is a structure-specific quality-control step whose necessity and sufficiency for prospective enrichment require systematic, multi-target validation.

Key words molecular docking redocking validation cryo-EM ATR kinase virtual screening AutoDock Vina BRICS CASF-2016

1 Introduction

Molecular docking is one of the most commonly employed computational tools in structure-based drug discovery because it provides a good compromise between speed and accuracy [1, 2]. Several factors are crucial for the reliability of the results obtained: receptor structure, ligand preparation, scoring function, search algorithm, and validation protocol. One of the most critical among these is the quality and conformation of the receptor structure, since all downstream analyses will inherit any errors introduced at this stage. This article aims to test that claim directly.

Cryo-EM has become an effective method for determining macromolecular structures, competing in many cases with X-ray crystallography at increasingly high resolution [3, 4]. Although resolution and a deposited structure's suitability for computational docking might be expected to track together, recent work indicates that medium-resolution cryo-EM structures, cross-docked under a standard protocol, succeed at only about half the rate of high-resolution crystal structures [5]. Several recent studies have also emphasized the importance of per-structure validation using AlphaFold2-predicted models [6, 7]: structure-specific empirical testing may be required regardless of how high-quality the input model appears, or its source.

ATR and CHK1 are ATP-competitive kinases with several chemical inhibitors reported for structure-based design [8–13]. Recently published cryo-EM structures of the ATR kinase domain (PDB 9L40, bound to berzosertib, 2.87 Å; PDB 9L4B, bound to camonsertib, 3.20 Å) [14] provide better starting points for docking than the original 4.7 Å ATR–ATRIP complex (PDB 5YZ0). In a companion study, berzosertib was shown to engage an additional dimer-interface allosteric site [14], further highlighting the intrinsic complexity of this system. The challenge posed by these data is practical, at the level of experimental protocol, rather than target biology: docking to a deposited cryo-EM coordinate set treats those coor-

dinates as the truth, but if that single snapshot model does not reproduce the ligand pose on which it was based, no amount of scoring innovation can restore a meaningful answer. We therefore asked whether nominal resolution and standard cryo-EM map-model quality metrics are sufficient to determine whether these deposited structures can reproduce their own experimental ligand poses under a fixed docking protocol.

Reproducing a receptor’s own bound ligand pose (cognate redocking) is an important sanity check for any docking methodology, but it does not necessarily indicate predictive capability for prospective screening [15]. Our work should be read as a methodological investigation into this process, not a broad virtual screening exercise. We also remind the reader that predicted binding, while useful, cannot provide evidence of functional modulation [8, 9].

2 Materials and methods

2.1 Receptor structures and preparation

Three receptor structures were retrieved in mmCIF format from the RCSB PDB AWS Open Data mirror (snapshot 2026-01-01): PDB 9L40 (ATR, bound to berzosertib, 2.87 Å), PDB 9L4B (ATR, bound to camonsertib, 3.20 Å), and PDB 2YM8 (human CHK1, 2.07 Å) [14]. We include 2YM8 as a non-cryo-EM (X-ray crystallographic) kinase reference structure to test whether our findings are reproducible on non-cryo-EM-derived receptors; with only one such structure to compare against, this can only be treated as a hypothesis ($n = 1$), which we report as such (Limitations). PDB 5YZ0 (4.7 Å) was omitted due to its poor resolution and consequent inability to reliably resolve side chains within the ATP pocket [14]. Both protein chains (full homodimer) were retained using Biopython [16].

For 9L40, contact analysis (heavy atoms within 4.5 Å of each protein chain) identified residues A2701/B2701 as the ATP-competitive active site and A2702/B2702 as the dimer-interface allosteric site [14]. The centroid-to-centroid distance between these sites is 60.65 Å, almost three times the 22 Å docking box used here, so the ATP-pocket redocking result reported below cannot have been affected by the allosteric site.

Receptor preparation was performed with Open Babel 3.1.1 [17] and Meeko (2025). For each structure:

- Hydrogens were added at pH 7.4 (Open Babel `-pH 7.4`);
- Alternate locations were resolved by keeping the highest-occupancy conformer; none of the alternate conformations are located near the cognate ligand in any of the three primary structures (within 5 Å);
- Crystallographic waters and non-ligand heteroatoms were removed;
- Asp/Glu negative; Lys/Arg positive; histidine tautomer states were determined by Meeko during the flexible-receptor preparation step, which requires an explicit HIE assignment for HIS2477 (Section 2.8);
- Meeko’s receptor-preparation step deleted ~400 residues whose side-chain density could not be matched against a template at approximately 3 Å resolution; automatically dropping surface/loop side chains that cannot be reliably placed in an incomplete cryo-EM map is a standard feature of receptor preparation at this resolution regime, not a study-specific deviation, and all deleted residues were >10 Å from the docking box (Supplementary Table S2);
- Proteins were converted to PDBQT with Open Babel using Gasteiger-Marsili partial charges [18] (`-gen3d -partialcharge gasteiger`).

All receptors and ligands in this study (redocking set, pilot library, CAK survey, and CASF-2016 subset alike) were docked at physiological pH 7.4 and using Gasteiger-Marsili charges. These are parameter-free, per-atom default charge models that do not need any target- or ligand-specific tweaking – the appropriate choice for a protocol that is deliberately held constant across a structurally and chemically diverse set of receptors and compounds rather than tailored specifically for any of them. A more subtle charge model or per-compound prediction of protonation state would reintroduce precisely the type of ligand-specific tweaking that this study’s design aims to avoid.

Because all ligands docked in this study are single-site, ATP-competitive kinase inhibitors of approximately 550 Da or less, the 22 Å cubic search box, centered on the cognate ligand centroid, was retained as the sole, unchanged protocol parameter for all receptors and all 45 library/survey ligands (5 seeds, 35 derivatives, and the CASF/CAK cognate ligands) instead of being sized per ligand: it is large enough to contain both the ligand itself and its first-shell pocket residues in every structure tested, which renders a static box size neutral with respect to the comparison between receptors. This follows the advice of Feinstein and colleagues [19] that search-box size be rationalized according to the specific ligand and receptor rather than chosen by default; no systematic per-ligand box-size sensitivity sweep across the full benchmark was undertaken, which is noted as a limitation (Section 3.7).

wwPDB validation reports give per-ligand information for the active-site ligand copy used as the redocking reference in each structure: 9L40/A2701, Q-score 0.553, residue inclusion 0.818; 9L4B/A2701, Q-score 0.528, residue inclusion 0.967 [3]. These values indicate that both deposited ligand models show similar local map support, although a similar Q-score does not guarantee that the pose itself is correct [3].

2.2 Seed library and derivatives

Five compounds comprised the seed library: the three deposited active-site ligands (berzosertib, camonsertib, and the CHK1 inhibitor from 2YM8), plus prexasertib and ceralasertib (SMILES checked against reported molecular formula and exact mass). These were split into 22 unique fragments using the BRICS algorithm [20], recombined using RDKit [21], and capped at 5,000 candidates. Candidates were pre-filtered by relaxed Lipinski [22] and Veber [23] rules, plus PAINS [24]/Brenk structural-alert removal [25], leaving 786 candidates. Retained compounds were assessed against four drug-likeness rules (Lipinski, Veber, Ghose [26], Egan [27]), of which 680 of 786 (86.5%) satisfied $\geq 3/4$ rules. Derivatives selected for pilot docking ($n = 35$) were those satisfying the most rules, ranked by ascending molecular weight. The complete ranking and selection scores are in Supplementary Table S4. This selection protocol is necessarily biased toward smaller, rule-compliant molecules, which should be kept in mind when interpreting the pilot docking ranking.

2.3 Redocking validation protocol

Each receptor was validated by extracting its deposited ligand, redocking into the same receptor/box using AutoDock Vina 1.2.7 (Python bindings) [1] exhaustively (exhaustiveness 32), seeded with three independent replicates (seeds 1000, 1001, 1002), and calculating the RMSD (heavy atoms) between the redocked pose and the deposited native pose, using a pre-specified pass threshold of ≤ 2.0 Å [2]. The threshold was set to conform to the Astex Diverse Set convention [2] and was pre-specified prior to testing any receptor, to prevent post-hoc threshold selection.

Two RMSD measures are reported: naive index-matched RMSD and symmetry-corrected RMSD (spyrmsd v0.6.2, graph-isomorphism atom matching [28]). The symmetry-corrected RMSD is defined as the root-mean-square deviation between a docked pose P_d and a reference pose P_r :

$$\text{RMSD}_s(P_d, P_r) = \min_{\sigma \in S} \sqrt{\frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_{d,i} - \mathbf{x}_{r,\sigma(i)}\|^2}$$

where σ is a permutation of equivalent atoms in the set S of topological symmetries and N is the number of heavy atoms. This correction is useful for compounds like berzosertib that contain internal topological symmetries (equivalent ring atoms, sulfone oxygens), for which the naive index-matched RMSD may overstate the true error of the pose [28].

Search reproducibility ρ_{repro} is the average pairwise RMSD between the best-scoring poses of each replicate across the three seeded runs:

$$\rho_{\text{repro}} = \frac{1}{3} \sum_{i < j} \text{RMSD}(P_{\text{best},i}, P_{\text{best},j})$$

This metric assesses the random variation in the Vina search rather than its accuracy against the native pose, a distinction kept separate throughout this study (Results).

Both sanity controls yielded the expected result, with crystal-vs-itself giving exactly 0.000 Å and the exact 1.0 Å rigid translation recovered as exactly 1.000 Å. If a receptor failed the redocking control in any replicate, it was excluded from candidate scoring.

2.4 Ligand preparation

For cognate redocking, the deposited ligand chemical state (connectivity, stereochemistry, protonation, tautomer) was preserved unchanged, since the purpose of that test is to ask whether the docking protocol can recover a known experimental pose, not to explore alternative protonation states. For the pilot screen, all seed and derivative ligands were prepared by 3D embedding, protonating at pH 7.4 with Open Babel, assigning Gasteiger-Marsili partial charges, and converting to PDBQT – the same parameter-free treatment used for every receptor and ligand in this study (Section 2.1). A more detailed per-compound search of tautomers/protomers was not attempted and is not claimed here.

2.5 Pilot library docking

The 5 seeds plus the 35 selected derivatives were docked against each passing receptor using the identical protocol except at exhaustiveness 16 (three seeds: 2000, 2001, 2002). For consistency of the protocol argument, we validated redocking at this screening-level exhaustiveness as well: the same pass/fail pattern held (9L40 3.01 ± 0.06 Å fail, 9L4B 0.59 ± 0.02 Å pass, 2YM8 0.92 ± 0.01 Å pass), indicating that the reliability conclusion is robust to this choice of parameter.

2.6 Multi-structure survey

The identical validation protocol was also applied to 14 additional cryo-EM structures of the CDK7/cyclin H/MAT1 CAK complex [29], each bound to a different ATP-competitive inhibitor. All 14 structures come from one target system and one research group, making this a clustered single-target set rather than 14 independent targets. Given this close relationship, the 13/14 result for this set should be viewed as descriptive evidence of failure within this series, not a population estimate. The pipeline was verified on 9L4B: it reproduced the original result almost exactly (RMSD 0.60 ± 0.03 vs. 0.61 ± 0.02 Å; Q-score 0.528, identical), confirming that the automated survey pipeline is consistent with the hand-curated protocol.

2.7 Positive-control sanity check

Twenty CASF-2016 [30] complexes were selected by an objective, non-random rule fixed before any of the 20 was inspected individually: the first 20 PDB IDs, in file order, of a complete CASF-2016 core-set listing redistributed, not curated or re-ordered by us, in the DeepDock GitHub repository.¹ This sequential rule, rather than a stratified or pseudorandom draw, was chosen specifically to avoid any appearance of hand-picking targets favorable to the pipeline. The 285-complex CASF-2016 core set [30] is a popular test case for docking protocol validation, and the reported 58% Vina self-docking success rate on this benchmark [31] is a helpful contextual comparison. That value was obtained under particular preparation and sampling conditions distinct from ours, so we do not take it as representative of any “Vina success rate.” This test is meant purely as a check on our pipeline, not as a broad benchmark of docking accuracy: 20 complexes, under one fixed generic protocol, cannot replace a systematic evaluation of accuracy, and failing to reject a null hypothesis of equality to the 58% literature value does not itself indicate equality. We report exact binomial 95% confidence intervals using the Clopper-Pearson method [32].

2.8 Flexible-sidechain redocking

As a test for any potential role of rigid conformations in the failure of 9L40, five active-site residues within 4.5 Å of native berzosertib (chain A: LYS2327, TYR2365, GLU2378, TRP2379, HIS2477) were selected *a priori* (not based on inspection of the redocked poses) and defined as flexible side chains for Vina. The 4.5 Å distance cutoff matches the same heavy-atom contact-shell radius used earlier in this study to differentiate the ATP-competitive site from the allosteric site (Section 2.1); the five identified residues are therefore a natural, predetermined consequence of applying this fixed radius to the native pose, not a deliberate choice for effect.

The flexible and rigid parts of the receptor were prepared with Meeko (`mk_prepare_receptor.py`), which requires an explicit tautomer assignment for every histidine, including HIS2477 (HIE). As in Section 2.1, template-mismatched surface residues were automatically deleted by Meeko; none of the five targeted flexible residues was among them. To give the flexible side chains room to explore alternative rotamers without triggering a ligand-outside-box error, the docking box size was increased from 22 Å (rigid redocking, Section 2.1) to 30 Å. Because this changes three things relative to the original rigid result at once – flexibility, box size, and the Meeko-prepared receptor – a matched control was run alongside the flexible run: fully rigid redocking (no residues declared flexible) with the identical enlarged 30 Å box and

the identical Meeko-prepared receptor and ligand, isolating the effect of flexibility from the other two changes. Both conditions used three seeded replicates (1000, 1001, 1002, exhaustiveness 32), consistent with the original rigid-redocking protocol. Individual replicate RMSDs are provided in Supplementary Table S5.

2.9 Rank concordance analysis

To provide a fair comparison, the 40-compound pilot set was docked against excluded 9L40 using the same three seeds as for 9L4B (2000, 2001, 2002). Rank concordance metrics included: Spearman rank correlation ρ with bootstrap 95% confidence interval (10,000 resamples), Kendall τ , mean absolute rank displacement, and top-10 overlap fraction. For ρ , the bootstrap approach estimates uncertainty by resampling compound pairs with replacement [33], accounting for the small set size.

2.10 Computational environment and reproducibility

All calculations: Vina 1.2.7 (Python bindings), Open Babel 3.1.1, RDKit 2026.03.6, Biopython 1.88, Meeko (2025), all CPU only on a standard workstation (no GPU acceleration). Exact command-line arguments, prepared PDBQT files, and individual replicate RMSDs are provided in the repository. Supplementary Table S6 provides the three seeded RMSD values for each structure individually. Reported mean $\pm$ SD describes variation among the three docking runs but is not an inferential measure of uncertainty about the receptor’s suitability. All software versions, random seeds, thresholds, and selection criteria were recorded prior to the first execution of the pipeline and are available for independent verification.

3 Results and discussion

3.1 Redocking validation

Redocking validation results are summarized in Table 1. 9L4B and 2YM8 passed comfortably (symmetry-corrected RMSD 0.61 ± 0.02 Å and 0.76 ± 0.01 Å). 9L40, the higher-resolution structure, failed (RMSD 2.53 ± 0.01 Å). The nearest allosteric-site ligand copy is 60.65 Å from the active site, nearly three times the 22 Å docking box, which rules out the possibility that the docking box reached the allosteric site. Per the pre-specified rule, 9L40 was excluded from the candidate-scoring process. Figure 1 shows the native and docked poses for both structures.

¹ <https://github.com/OptiMaL-PSE-Lab/DeepDock>, file `Validation_Docking/DockingResults_CASF2016_CoreSet.csv`.

A: 9L40 (berzosertib), FAILED, RMSD 2.53 Å

B: 9L4B (camonsertib), PASSED, RMSD 0.61 Å

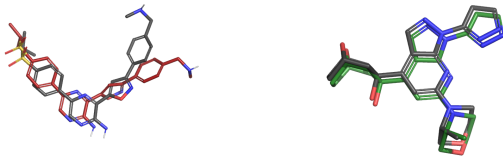


Fig. 1: Native (grey) vs. redocked (colored) best pose. Left: 9L40/berzosertib (symmetry-corrected RMSD 2.53 Å). Right: 9L4B/camonsertib (0.61 Å).

Table 1: Redocking validation (three seeded Vina replicates, exhaustiveness 32)

Structure	Res. (Å)	RMSD (Å)	Pass?	Repro. (Å) ^a
9L40 (ATR)	2.87	2.53±0.01	No	0.56
9L4B (ATR)	3.20	0.61±0.02	Yes	0.045
2YM8 (CHK1) ^b	2.07	0.76±0.01	Yes	0.078

^a Search reproducibility: mean pairwise RMSD between replicate best poses. ^b Non-cryo-EM (X-ray) reference structure.

3.2 Multi-structure survey

Table 2 reports results for all 17 structures tested. Of these, 14 of 17 (82%) do not pass the 2.0 Å threshold; only three pass it: 9L4B, 2YM8, and one CAK entry (8P75). There was no statistically significant monotonic association between reported resolution and redocking outcome (Spearman $\rho = -0.28$, $p = 0.28$, $n = 17$), nor between the evaluated global measure of residue inclusion and redocking outcome ($r = -0.14$, $p = 0.61$, $n = 16$). For the 12 structures with a reported Q-score, the two passing structures had a lower mean Q-score (0.569) than the 10 failing structures (0.688), which implies a descriptive trend but is not a definitive statistical correlation given the small number of Q-scored passing structures ($n = 2$); this result should not be taken as an indication that Q-score has no predictive value.

The CAK failures showed RMSDs in the range 5–7 Å but docked ligand centroids within 1.1–4.1 Å of their native centroids, suggesting the docked ligand had been reoriented within the pocket rather than moved far away from it. This is consistent with orientational degeneracy at the kinase hinge – where multiple positions may be reasonable for an ATP-competitive scaffold – or with rigid-receptor limitations that prevent the docking engine from exploring the experimentally observed binding mode, rather than with a setup problem such as a misplaced box.

The 14 CAK structures come from a single target system and research group; they are thus clustered observations, not 14 independent targets. Within this single-target series, 13 of 14 structures failed the prespecified

Table 2: Multi-structure survey, all 17 structures tested, sorted by symmetry-corrected RMSD. n/a: Q-score not reported for this experimental method (X-ray) or ligand occupancy state (split-altloc CAK ligands).

Structure	Res. (Å)	RMSD (Å)	Pass?	Q-score	Res. incl.
9L4B (ATR)	3.20	0.61	Yes	0.528	0.967
2YM8 (CHK1)	2.07	0.76	Yes	n/a	n/a
8P75 (CAK)	2.00	1.03	Yes	0.609	0.724
9L40 (ATR)	2.87	2.53	No	0.553	0.818
8P72 (CAK)	1.90	2.70	No	0.707	0.963
8P78 (CAK)	1.90	2.98	No	n/a	0.828
8P6Z (CAK)	2.10	4.84	No	0.705	0.885
8P77 (CAK)	1.80	5.02	No	n/a	0.793
8P73 (CAK)	2.00	5.84	No	0.722	0.800
8P6V (CAK)	1.90	6.14	No	0.738	0.966
8P7L (CAK)	2.10	6.16	No	0.708	0.969
8P70 (CAK)	2.00	6.25	No	n/a	1.000
8P76 (CAK)	2.00	6.60	No	0.640	0.735
8P6X (CAK)	1.90	6.60	No	n/a	0.964
8P74 (CAK)	2.20	6.67	No	0.628	0.793
8P71 (CAK)	2.00	7.02	No	0.720	0.926
8P6W (CAK)	1.90	7.33	No	0.762	0.929

criterion. Because of this clustering, the 13/14 observation should not be interpreted as an estimate of the underlying failure probability for kinase structures generally, but rather as descriptive evidence of failure within this particular CAK series.

3.3 Positive-control sanity check

The unmodified pipeline passed 8 of 20 (40%) CASF-2016 subset complexes selected by the first-20-in-file-order rule described in Methods (Table 3). The exact 95% binomial confidence interval (Clopper-Pearson) is 19.1–63.9%. We note that the purpose of this subset is as a pipeline sanity check, not an exhaustive evaluation of docking accuracy: failure to reject a null hypothesis does not suggest equivalence, only that we are unable to detect a significant difference. It indicates that our pipeline is not badly broken, but not that our pipeline is definitively valid. Although the observed 40% is not significantly different from the literature Vina value of 58% on the full 285-complex core set [31] (one-sided binomial $p = 0.08$), the comparison has caveats, particularly the small sample size and differences in software version, receptor preparation, and sampling parameters between our protocol and the one the 58% figure was measured under. The 58% value reflects Vina performance in a specific randomized-ligand redocking experiment and should not be construed as a general “Vina success rate,” as other CASF-2016 evaluations using different protocols report substantially higher top-pose success rates (Supplementary Section S1).

The CAK series (1/14 passed, 7.1%; exact 95% CI 0.2–33.9%) showed fewer successful redockings than the

Table 3: Summary of redocking outcomes

Set	<i>n</i>	Pass (rate)	95% CI
CASF-2016 subset (pipeline sanity check)	20	8 (40%)	19.1–63.9%
CAK series (single-target)	14	1 (7.1%)	0.2–33.9%
ATR pair (9L40/9L4B)	2	1 (50%)	1.3–98.7%

CASF sanity-check subset (8/20, 40%; Table 3). Fisher’s exact test gave $p = 0.037$ for the prespecified one-sided comparison (hypothesis: CAK would succeed less often than the CASF control) and $p = 0.050$ for the two-sided comparison. The CASF sanity check did not reveal an obvious gross failure of the docking implementation, whereas 13 of 14 structures in the sampled CAK series *failed* the prespecified criterion.

3.4 Matched flexible-sidechain analysis

Allowing flexibility in the five active-site residues nearest the native berzosertib pose did not recover it: mean symmetry-corrected RMSD across three seeds was 3.00 ± 0.92 Å (individual seeds: 4.29, 2.23, 2.49 Å), and no seed met the 2.0 Å threshold. Naively compared to the original fully rigid result (2.53 Å, 22 Å box, Open Babel-prepared ligand), it appears that allowing flexibility slightly worsens the pose – but this comparison is confounded, since the flexible run also changes the box size (22 Å → 30 Å) and the receptor preparation (Meeko vs. Open Babel) at the same time as flexibility. The matched control isolates flexibility alone: fully rigid redocking with the identical enlarged 30 Å box and the identical Meeko-prepared receptor and ligand, but no residues declared flexible, gives mean symmetry-corrected RMSD 4.50 ± 0.05 Å – worse than both the original rigid result and the flexible result. Against this correctly matched baseline, the improvement from 4.50 Å to 3.00 Å when the five residues are allowed to move is real and in the expected direction, though still insufficient to reach the 2.0 Å threshold. The larger box size and the Meeko receptor preparation, not the choice to keep side chains rigid, account for most of the difference between the original 2.53 Å result and either Meeko-based result. The corrected conclusion is therefore narrower than either the original claim or its opposite: flexibility genuinely helps, but not enough on its own to rescue the pose, so a less-supported deposited pose or a conformational change beyond these five side chains remains at least as plausible an explanation (Discussion).

3.5 Error-propagation comparison

Docking the 40-compound pilot set against the excluded 9L40 instead of validated 9L4B produced low rank concordance: Spearman $\rho = 0.21$ (bootstrap 95% CI -0.19

Table 4: Largest rank shifts between 9L40 and 9L4B scoring

Compound	Rank (9L40)	Rank (9L4B)	Shift
deriv_017	5	34	29
deriv_010	10	39	29
deriv_022	32	7	25
deriv_033	34	9	25

to 0.52, crossing zero), Kendall $\tau = 0.09$, mean absolute rank displacement 11.7 positions (max 29), and only 5 of 10 compounds shared between the two top-10 lists (Table 4). The five shared top-10 compounds were: camonsertib, ceralasertib, deriv_002, deriv_003, and deriv_019 – the same five compounds that occupied the top five ranks in the 9L4B ranking. The top of the ranking is therefore stable across receptors: a campaign selecting only its top five candidates for synthesis would reach the same set either way. The churn is concentrated in the mid-ranks (roughly 6–40), where score gaps are well below Vina’s commonly cited absolute error against experiment (2–3 kcal/mol); a bootstrap ρ CI that crosses zero describes substantial mid-rank reshuffling, not a scrambled ranking overall. A reasonable interpretation is that receptor choice alters the rankings of the middle portion of this pilot screen but does not affect the highest ranks: a selection from the top five (or top ten) is relatively insensitive to receptor choice, while a selection from the top twenty (or top thirty) would draw on a materially different set of compounds. Again, this is a demonstration of consequence within this pilot set, not a general enrichment claim, which would require evaluation against known actives and decoys [34].

3.6 Implications for practice

Cognate redocking is a structure-specific quality-control test that can catch docking errors not apparent from nominal resolution or the evaluated global map/model metrics. 9L40 and 9L4B were solved in the same study, yet only the lower-resolution one passed – a flip that no proxy metric detected. This observation was reproduced in a separate CAK structure series, though the CAK structures come from a single target system and so do not establish generality independent of target. The 13/14 observed failures in the sampled CAK series should be regarded as descriptive evidence within that series, not as a kinase-wide failure frequency.

The small size of this survey means that any association seen between the available global map/model metrics and redocking outcome should not be over-interpreted. Only two structures with available Q-score measurements passed, so the comparison of the Q-score group is descriptive rather than definitive and should not be generalized beyond this sample. The shortcut to skip the redocking-

validation step this paper argues against is not a neutral one either: the error-propagation comparison above shows that scoring candidates against an unvalidated receptor does reorder a screen’s mid-ranks even where it preserves its very top, so the real-world cost of skipping validation depends on how far down a ranked list a downstream campaign picks candidates. While these results provide evidence for using per-structure cognate redocking as a quality-control step, the systematic multi-target evaluation needed to decide on its general utility has yet to be done.

3.7 Limitations

This is a small pilot ($n = 17$). The CAK set is a single-target convenience sample, not a systematic survey of recent kinase depositions. The global Q-score analysis is underpowered (2 passing structures with reported Q-scores). Local density-fit metrics were assessed for 9L40 and 9L4B but a full per-residue active-site analysis across all 17 structures was not performed, as such an analysis would require uniform local validation-report data that were not consistently available for the CAK series. No molecular dynamics, MM-GBSA, or cross-kinase selectivity panel was run. The BRICS derivative library is small (5 seeds, 35 derivatives) and biased toward smaller, rule-compliant molecules; enrichment-based screening metrics cannot be meaningfully computed at this scale. Most importantly, nothing in this study establishes functional modulation—only predicted binding and ranking.

4 Conclusions

Cognate redocking is a structure-specific quality-control test that can reveal failures not obvious from nominal resolution or the evaluated global map/model metrics. Applying this test to two ATR structures from the same study showed success on the lower-resolution structure and failure on the higher-resolution one, an inversion not explained by allosteric-site proximity. A single-target CAK structure set showed the cognate-redocking criterion failing in 13 of 14 instances. Its value and utility for prospective enrichment nonetheless need further assessment across a wider, multi-target benchmark. Omitting this step is not innocuous, as shown by this pilot’s own error-propagation comparison, which indicates that scoring candidates against an unvalidated receptor reorders a screen’s mid-ranks, with the degree of impact depending on how far down a subsequent campaign’s candidate selection goes – even in this pilot, where the very top ranks happened to be preserved. The next logical step is to test whether this proportion of failures holds for recently published kinase structures more broadly, through a systematic study of many independent targets. The code, data, and structures behind this work are all available online for independent verification.

Acknowledgements

[Placeholder]

Author contributions

CRedit roles (E.Y.: Ethan Ye; A.P.: Aryan Padarthi; S.K.: Sungju Kim; A.E.C.: Alexandre Elie-Dit-Cosaque; E.J.: Eunho Jo). Draft assignment, to be confirmed by all authors prior to submission: Conceptualization – E.Y.; Methodology – E.Y.; Software – E.Y.; Validation – E.Y., A.P., S.K., A.E.C., E.J.; Formal analysis – E.Y.; Investigation – E.Y., A.P., S.K., A.E.C., E.J.; Data curation – E.Y.; Writing – original draft – E.Y.; Writing – review & editing – E.Y., A.P., S.K., A.E.C., E.J.

Data availability

All code, seed and derivative compound tables, redocking validation results, individual replicate RMSDs, prepared PDBQT files, and full residue-deletion lists are available at <https://github.com/ethanyyy123/research-molecule-editing> and in the underlying analysis pipeline repository at <https://github.com/The-MathKing/CogRed>, which also tracks every review round and correction described in this manuscript. Receptor structures are publicly available from RCSB PDB (9L40, 9L4B, 2YM8, and the CDK7/cyclin H/MAT1 and CASF-2016 structures cited in Tables 2–3) and were retrieved via the AWS Open Data mirror ([s3://pdxnlp](https://pdxnlp.github.io/pdxnlp/)).

Declarations

Conflict of interest: The authors declare no competing interests.

Ethical approval: Not applicable. This study is a purely computational analysis of publicly deposited structures and did not involve human participants, human data, or animal subjects.

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References

- Oleg Trott and Arthur J. Olson. Autodock vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of Computational Chemistry*, 2010. doi: 10.1002/jcc.21334.
- Michael J. Hartshorn, Marcel L. Verdonk, Gianni Chessari, Suzanne C. Brewerton, Wijnand T. M. Mooij, Paul N. Mortenson, and Christopher W. Murray. Diverse, high-quality test set for the validation of protein-ligand docking performance. *Journal of Medicinal Chemistry*, 50(4):726–741, 2007. doi: 10.1021/jm061277y.
- Grigore Pintilie, Kai Zhang, Zhaoming Su, Shiheng Li, Michael F. Schmid, and Wah Chiu. Measurement of atom resolvability in cryo-em maps with q-scores. *Nature Methods*, 17:328–334, 2020. doi: 10.1038/s41592-020-0731-1.
- Werner Kühlbrandt. The resolution revolution. *Science*, 343(6178):1443–1444, 2014. doi: 10.1126/science.1251652.
- Seho Lee et al. Benchmarking applicability of medium-resolution cryo-EM protein structures for structure-based drug design. *Journal of Computational Chemistry*, 44(14):1360–1368, 2023. doi: 10.1002/jcc.27091.
- Matthew Holcomb, Ya-Ting Chang, David S. Goodsell, and Stefano Forli. Evaluation of AlphaFold2 structures as docking targets. *Protein Science*, 32(1):e4530, 2023. doi: 10.1002/pro.4530.
- Valeria Scardino, Juan I. Di Filippo, and Claudio N. Cavasotto. How good are AlphaFold models for docking-based virtual screening? *iScience*, 26(1):105920, 2022. doi: 10.1016/j.isci.2022.105920.
- Akiko Kumagai, Jean Lee, Hae Yong Yoo, and William G. Dunphy. Topbp1 activates the atr-atrip complex. *Cell*, 2006.
- Trent E. Bass, Jessica W. Luzwick, Gina Kavanaugh, et al. Etaa1 acts at stalled replication forks to maintain genome integrity. *Nature Cell Biology*, 2016.
- Joshua C. Saldivar, David Cortez, and Karlene A. Cimprich. The essential kinase atr: ensuring faithful duplication of a challenging genome. *Nature Reviews Molecular Cell Biology*, 2017.
- Anne M. Casper, Paul Nghiem, Michael F. Arlt, and Thomas W. Glover. Atr regulates fragile site stability. *Cell*, 2002.
- Michelle Debatisse, Benoit Le Tallec, Anne Letessier, Bernard Dutrillaux, and Olivier Brison. Common fragile sites: mechanisms of instability revisited. *Trends in Genetics*, 2012.
- Emmanouil Fokas, Remko Prevo, Jonathan R. Pollard, et al. Targeting atr in vivo using the novel inhibitor ve-822 results in selective sensitization of pancreatic tumors to radiation. *Cell Death & Disease*, 2012.
- Xuejuan Wang et al. Molecular architecture and inhibition mechanism of human atr-atrip. *Science Bulletin*, 70(13):2137–2146, 2025. doi: 10.1016/j.scib.2025.05.009.
- Ajay N. Jain and Anthony Nicholls. Recommendations for evaluation of computational methods. *Journal of Computer-Aided Molecular Design*, 22:133–139, 2008. doi: 10.1007/s10822-008-9196-5.
- Peter J. A. Cock, Tiago Antao, Jeffrey T. Chang, et al. Biopython: freely available python tools for computational molecular biology and bioinformatics. *Bioinformatics*, 2009.
- Noel M. O’Boyle, Michael Banck, Craig A. James, Chris Morley, Tim Vandermeersch, and Geoffrey R. Hutchison. Open babel: An open chemical toolbox. *Journal of Cheminformatics*, 2011.
- Johann Gasteiger and Mario Marsili. Iterative partial equalization of orbital electronegativity—a rapid access to atomic charges. *Tetrahedron*, 36(22):3219–3228, 1980. doi: 10.1016/0040-4020(80)80168-2.
- Wei P. Feinstein and Michal Brylinski. Calculating an optimal box size for ligand docking and virtual screening against experimental and predicted binding pockets. *Journal of Cheminformatics*, 7:18, 2015. doi: 10.1186/s13321-015-0067-5.
- Jörg Degen, Christoph Wegscheid-Gerlach, Andrea Zaliani, and Matthias Rarey. On the art of compiling and using ‘drug-like’ chemical fragment spaces. *ChemMedChem*, 2008.
- RDKit. Rdkit: Open-source cheminformatics. <http://www.rdkit.org>, 2024.
- Christopher A. Lipinski, Franco Lombardo, Beryl W. Dominy, and Paul J. Feeney. Experimental and computational approaches to estimate solubility and permeability in drug discovery and development settings. *Advanced Drug Delivery Reviews*, 23:3–25, 1997.
- Daniel F. Veber, Stephen R. Johnson, Hung-Yuan Cheng, et al. Molecular properties that influence the oral bioavailability of drug candidates. *Journal of Medicinal Chemistry*, 2002.
- Jonathan B. Baell and Georgina A. Holloway. New substructure filters for removal of pan assay interference compounds (pains). *Journal of Medicinal Chemistry*, 2010.
- Ruth Brenk, Alessandro Schipani, Daniel James, Agnes Krasowski, Ian H. Gilbert, Julie Frearson, and Paul G. Wyatt. Lessons learnt from assembling screening libraries for drug discovery for neglected diseases. *ChemMedChem*, 3(3):435–444, 2008. doi: 10.1002/cmdc.200700139.
- Arup K. Ghose, Vellarkad N. Viswanadhan, and John J. Wendoloski. A knowledge-based approach in designing combinatorial or medicinal chemistry libraries for drug discovery. *Journal of Combinatorial Chemistry*, 1999.

27. William J. Egan, Kenneth M. Merz, and John J. Baldwin. Prediction of drug absorption using multivariate statistics. *Journal of Medicinal Chemistry*, 2000.
28. Rocco Meli and Philip C. Biggin. spyrmsd: symmetry-corrected rmsd calculations in python. *Journal of Cheminformatics*, 12:49, 2020. doi: 10.1186/s13321-020-00455-2.
29. Victoria I. Cushing, Ai Fang Koh, Jing Feng, Kotryna Jurgaityte, Alexander Bondke, Simon H. B. Kroll, Maud Barbazanges, Bernhard Scheiper, Anup K. Bahl, Anthony G. M. Barrett, Simak Ali, Abhay Kotecha, and Basil J. Greber. High-resolution cryo-em of the human cdk-activating kinase for structure-based drug design. *Nature Communications*, 15:2265, 2024. doi: 10.1038/s41467-024-46375-9.
30. Minyi Su, Qifan Yang, Yu Du, Guoqin Feng, Zhihai Liu, Yan Li, and Renxiao Wang. Comparative assessment of scoring functions: The CASF-2016 update. *Journal of Chemical Information and Modeling*, 59(2):895–913, 2019. doi: 10.1021/acs.jcim.8b00545.
31. Ute F. Röhrig, Maïte Goullieux, Matthieu Bugnon, and Vincent Zoete. Attracting cavities 2.0: Improving the flexibility and robustness for small-molecule docking. *Journal of Chemical Information and Modeling*, 63(12):3925–3940, 2023. doi: 10.1021/acs.jcim.3c00054.
32. C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. doi: 10.1093/biomet/26.4.404.
33. Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall/CRC, New York, 1993.
34. Niu Huang, Brian K. Shoichet, and John J. Irwin. Benchmarking sets for molecular docking. *Journal of Medicinal Chemistry*, 49(23):6789–6801, 2006. doi: 10.1021/jm0608356.